But their gravities were too strong to populate the Oort Cloud: Ice worlds that came close to them were gravitationally pitched out of the Solar System entirely—destined to wander forever in the great dark between the stars.

So the lovely comets that on occasion rouse us humans to wonder and to awe, that crater the surfaces of inner planets and outer moons, and that now and then endanger life on Earth would be unknown and unthreatening had Uranus and Neptune not grown to be giant worlds four and a half billion years ago.

THIS IS THE PLACE for a brief interlude on planets far beyond Neptune and Pluto, planets of other stars.

Many nearby stars are surrounded by thin disks of orbiting gas and dust, often extending to hundreds of astronomical units (AU) from the local star (the outermost planets, Neptune and Pluto, are about 40 AU from our Sun). Younger Sun-like stars are much more likely to have disks than older ones. In some cases, there's a hole in the center of the disk as in a phonograph record. The hole extends out from the star to perhaps 30 or 40 AU. This is true, for example, for the disks surrounding the stars Vega and Epsilon Eridani. The hole in the disk surrounding Beta Pictoris extends to only 15 AU from the star. There is a real possibility that these inner, dust-free zones have been cleaned up by planets that recently formed there. Indeed, this sweeping-out process is predicted for the early history of our planetary system. As observations improve, perhaps we will see telltale details in the configuration of dust and dust-free zones that will indicate the presence of planets too small and dark to be seen directly. Spectroscopic data suggest that these disks are churning and that matter is falling in on the central stars—perhaps from comets formed in the disk, deflected by the unseen planets, and evaporating as they approach too close to the local sun.

Because planets are small and shine by reflected light, they tend to be washed out in the glare of the local sun. Nevertheless, many efforts are now under way to find fully formed planets around nearby stars—by detecting a faint brief dimming of starlight as a